Lab 01:
Frequency Response of Operational Amplifiers

ECE 1212: Electronic Circuit Design Lab

Fall 2026

Stefan Zhang.....Student ID# 2024141520349
Nick Zhu .....Student ID# 2024141520329

Team Roles:

Activity	**Student Name**
Circuit Construction	Stefan Zhang
Data Collection	Nick Zhu
Data Analysis	Stefan Zhang; Nick Zhu
Report Writing	Stefan Zhang; Nick Zhu

Date Submitted: September 16, 2026

1) Objectives

This lab investigated the frequency response of the LM741 operational amplifier. The main goals were to:

- measure the 3-dB bandwidth of non-inverting amplifiers with $A_V = 1$ and $A_V = 10$;
- estimate the unity-gain bandwidth f_t;
- build a high-gain inverting amplifier with $R_{in} \approx 1\,\mathrm{k\Omega}$ and $f_{3\mathrm{dB}} \geq 20\,\mathrm{kHz}$;
- test DC compensation and the frequency dependence of input impedance.

For the LM741, the closed-loop bandwidth is approximately

$$A_V f_{3\mathrm{dB}} \approx f_t,$$

while for an inverting amplifier,

$$f_{3\mathrm{dB}} \approx \frac{f_t}{1 + R_f/R_{in}}.$$

2) Equipment Used

Table 1: Equipment and components used in Lab 01

Item	**Quantity**
LM741 operational amplifier	1 or more
Dual DC power supply ($\pm 12\,\mathrm{V}$)	1 set
Function generator	1
Digital oscilloscope	1
Digital multimeter	1
Breadboard and wires	1 set
Resistors: $1\,\mathrm{k\Omega}$, $3\,\mathrm{k\Omega}$, $27\,\mathrm{k\Omega}$, $47\,\mathrm{k\Omega}$, $3.3\,\mathrm{k\Omega}$	as needed
$10\,\mathrm{k\Omega}$ potentiometer	1
Series resistor R_s for input-impedance measurement	1

3) Part I: Non-Inverting Amplifiers

3.1) $A_V = 1$ Voltage Follower

The first circuit was a voltage follower with ideal gain

$$A_V = 1.$$

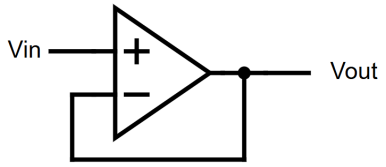


Figure 1: Non-inverting amplifier with $A_V = 1$

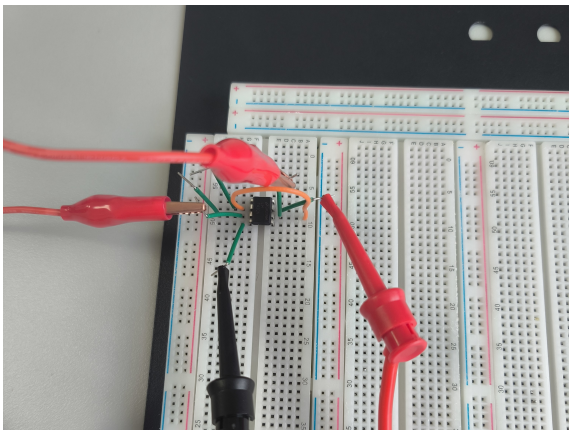


Figure 2: Experimental setup for the $A_V = 1$ amplifier

The measured 3-dB bandwidth was

$$f_{3\text{dB}} = 1.05\,\text{MHz}.$$

Thus,

$$f_t \approx A_V f_{3\text{dB}} = 1.05\,\text{MHz}.$$

This is about 5% above the nominal 1.0 MHz value.

3.2) $A_V = 10$ Non-Inverting Amplifier

For a non-inverting amplifier,

$$A_V = 1 + \frac{R_2}{R_1}.$$

Using

$$R_1 = 3\,\text{k}\Omega, \qquad R_2 = 27\,\text{k}\Omega,$$

gives

$$A_V = 10.$$

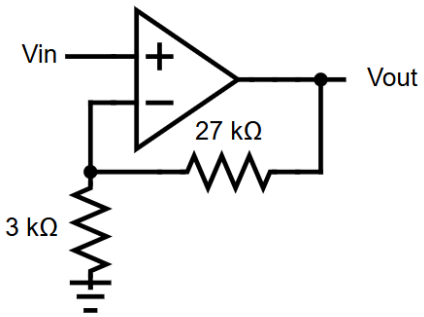


Figure 3: Non-inverting amplifier with $A_V = 10$

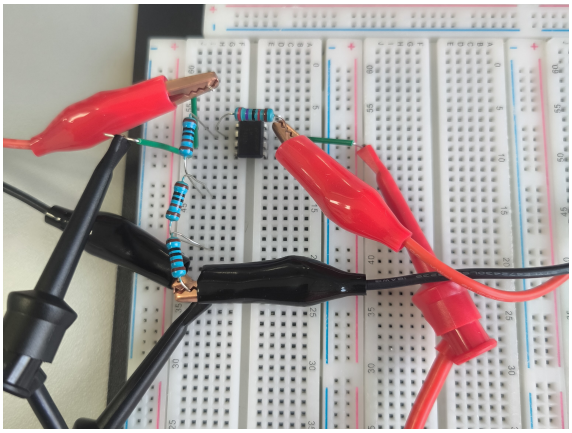


Figure 4: Experimental setup for the $A_V = 10$ amplifier

The measured 3-dB bandwidth was

$$f_{3\text{dB}} = 92\,\text{kHz}.$$

Therefore,

$$f_t \approx 10(92\,\text{kHz}) = 0.920\,\text{MHz}.$$

The average of the two measured estimates is

$$\overline{f_t} = 0.985\,\text{MHz},$$

which is close to the nominal LM741 value. The recorded input amplitude was approximately

$$V_{in} = 260\,\text{mV}_{pp}.$$

Table 2: Measured frequency response for $A_V = 10$

Frequency	V_{out}	$\lvert A_V \rvert$
100 kHz	1.720 V_{pp}	6.62
150 kHz	1.280 V_{pp}	4.92
200 kHz	0.940 V_{pp}	3.62
250 kHz	0.760 V_{pp}	2.92
300 kHz	0.632 V_{pp}	2.43

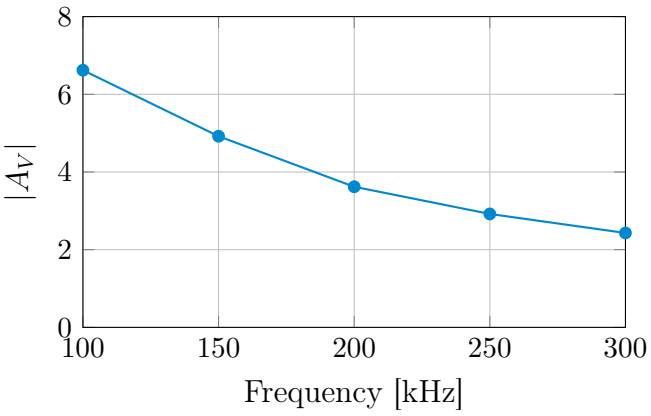


Figure 5: Frequency response for $A_V = 10$

3.3) Input Impedance

A series resistor R_s was used with the amplifier input as a voltage divider:

$$Z_{in} = R_s \frac{V_{in}}{V_s - V_{in}}.$$

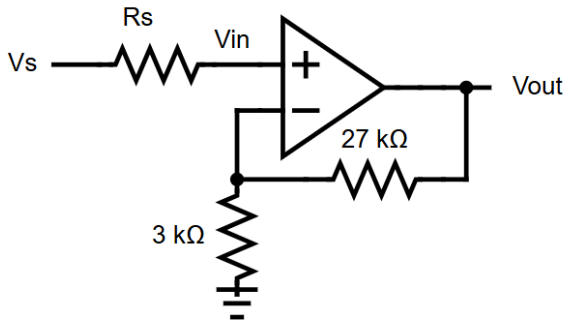


Figure 6: Input-impedance measurement circuit

Because these values are very large, oscilloscope loading can significantly affect the measurement.

Table 3: Input-impedance measurements at 1 kHz

Gain	R_s	V_s	V_{in}	Z_{in}
$A_V = 10$	1 MΩ	260.0 mV	144.0 mV	1.24 MΩ

4) Part II: High-Gain Inverting Amplifier

The design requirements were

$$R_{in} = 1\,\text{k}\Omega, \qquad R_L = 3.3\,\text{k}\Omega, \qquad f_{3\text{dB}} \geq 20\,\text{kHz}.$$

Using

$$R_f = 47\,\text{k}\Omega,$$

the nominal gain is

$$A_V = -\frac{R_f}{R_{in}} = -47.$$

The noise gain is

$$A_{\text{NG}} = 1 + \frac{R_f}{R_{in}} = 48.$$

Using $f_t = 1.05\,\text{MHz}$,

$$f_{3\text{dB}} \approx \frac{1.05\,\text{MHz}}{48} = 21.9\,\text{kHz}.$$

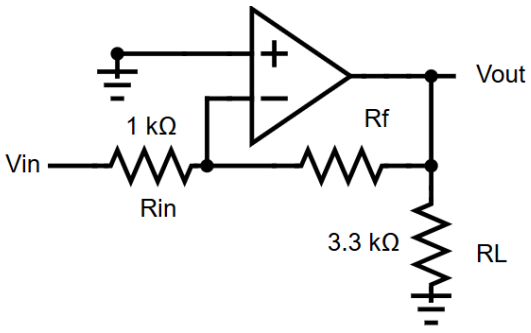


Figure 7: High-gain inverting amplifier

The recorded input amplitude was

$$V_{in} = 10.8\,\text{mV}.$$

The expected low-frequency output is

$$V_{out,0} \approx 47(10.8\,\text{mV}) = 507.6\,\text{mV}.$$

Thus the 3-dB level is

$$V_{out,3\text{dB}} = \frac{507.6}{\sqrt{2}} \approx 359\,\text{mV}.$$

Table 4: Measured output near the 3-dB frequency

Frequency	V_{out}
18 kHz	392 mV
19 kHz	384 mV
20 kHz	376 mV
21 kHz	368 mV
22 kHz	360 mV
23 kHz	352 mV
24 kHz	344 mV
25 kHz	330 mV
26 kHz	328 mV

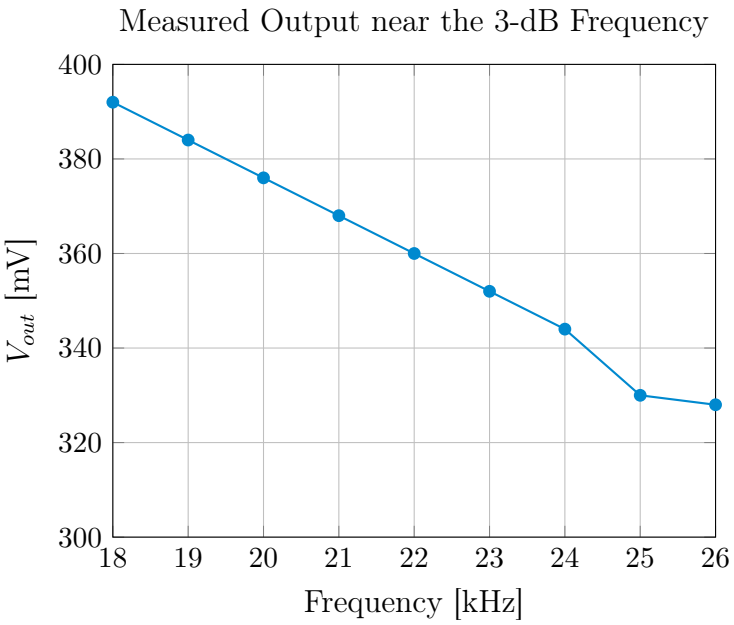


Figure 8: Output voltage near the 3-dB frequency

At 22 kHz,

$$V_{out} \approx 360\,\text{mV},$$

which is essentially the 3-dB level. Therefore,

$$f_{3\mathrm{dB}} \approx 22\,\mathrm{kHz}.$$

5) Part III: DC Compensation

A DC offset at the input is also amplified and can drive the output into saturation. A compensation voltage was therefore applied through

$$R_{\mathrm{comp}} = 1\,\mathrm{k\Omega}.$$

For DC,

$$V_{out,DC} = -R_f \left(\frac{V_{in,DC}}{R_{in}} + \frac{V_{\mathrm{comp}}}{R_{\mathrm{comp}}} \right).$$

Setting $V_{out,DC} = 0$ gives

$$V_{\mathrm{comp}} = -\frac{R_{\mathrm{comp}}}{R_{in}} V_{in,DC}.$$

Since

$$R_{\mathrm{comp}} = R_{in},$$

$$V_{\mathrm{comp}} = -V_{in,DC}.$$

Table 5: DC-compensation measurements

$V_{in,DC}$	V_{comp}	$V_{out,DC}$
1.00 V	−0.98 V	≈ 0 V

6) Part IV: Input Impedance of the Inverting Amplifier

The input impedance was measured using

$$Z_{in} = R_s \frac{V_{in}}{V_s - V_{in}}.$$

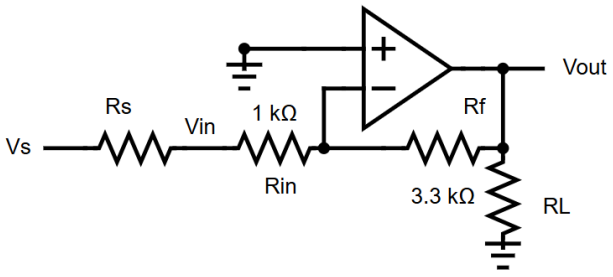


Figure 9: Input-impedance measurement circuit for the inverting amplifier

At low frequency,

$$Z_{in} \approx R_{in} = 1\,\mathrm{k}\Omega.$$

With finite open-loop gain,

$$Z_{in}(jf) \approx R_{in} + \frac{R_f}{1 + A(jf)}.$$

As frequency increases, $A(jf)$ decreases, so the virtual-ground approximation becomes weaker and Z_{in} deviates from $1\,\mathrm{k}\Omega$.

Table 6: Measured input impedance of the inverting amplifier

Frequency	**Measured** $\lvert Z_{in}\rvert$
100 Hz	1.03 kΩ
1 kHz	1.03 kΩ
5 kHz	1.03 kΩ
15 kHz	1.06 kΩ
20 kHz	1.17 kΩ

Measured Input Impedance of the Inverting Amplifier

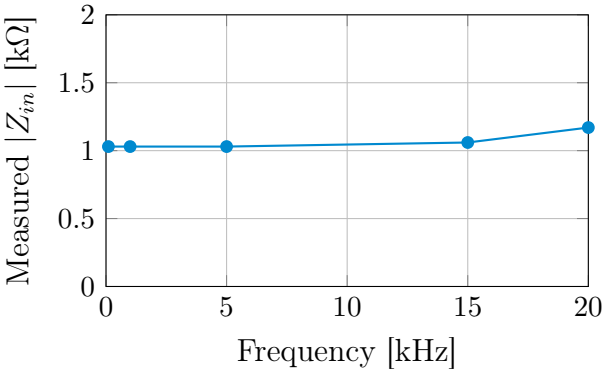


Figure 10: Measured input impedance versus frequency

7) Discussion

The measured results agree with the expected LM741 behavior. The voltage follower had a bandwidth of 1.05 MHz, while the $A_V = 10$ amplifier had a much smaller bandwidth of 92 kHz. The corresponding unity-gain bandwidth estimates were 1.05 MHz and 0.920 MHz, giving an average of 0.985 MHz.

For the inverting amplifier, the bandwidth is determined by the noise gain of 48. The predicted bandwidth was 21.9 kHz, and the measured 3-dB point was approximately 22 kHz, showing close agreement.

The non-inverting input impedance is very large, so probe and oscilloscope loading can affect the result. For the inverting circuit, the input impedance is close to $1\,\mathrm{k}\Omega$ at low frequency but changes as the open-loop gain decreases with frequency.

The DC-compensation network cancels the input DC offset and prevents output saturation, allowing the AC signal to remain within the available output range.

8) Conclusion

The experiment verified the gain-bandwidth behavior of the LM741. The measured values were

$$f_{3\text{dB}} = 1.05\,\text{MHz} \quad (A_V = 1),$$

$$f_{3\text{dB}} = 92\,\text{kHz} \quad (A_V = 10),$$

and

$$\overline{f_t} \approx 0.985\,\text{MHz}.$$

The high-gain inverting amplifier used

$$R_{in} = 1\,\text{k}\Omega, \qquad R_f = 47\,\text{k}\Omega,$$

giving

$$A_V = -47.$$

Its measured 3-dB bandwidth was approximately

$$22\,\text{kHz},$$

which satisfies the required 20 kHz bandwidth.

Overall, the measurements confirm that increasing closed-loop gain reduces bandwidth, and that the finite open-loop gain of the LM741 also affects input impedance and DC behavior.

References

[1] ECE 1212 Electronic Circuit Design Laboratory, *Experiment #1: Frequency Response of Operational Amplifiers*, Fall 2026.

[2] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed., Oxford University Press, 2015.

[3] Texas Instruments, *LM741 Operational Amplifier Datasheet*.